



CVM ANALYTICS CASE · CELLMOV CROSS-SELL

AI-Driven Offer Personalization: From One-Size-Fits-All to 4.7× Campaign Margin

Building an offer-recommendation model to optimize a CVM campaign with Machine Learning



EXECUTIVE SUMMARY

An AI offer recommender multiplies campaign margin 4.7× vs. today's fixed low-cost strategy, adding \$105K per 50K-lead wave



Situation

CellMov leads mobile but is a distant #2 in fixed broadband. Cross-sell via outbound calls is the growth lever, and the first offer pitched decides the call.



Complication

Offer assignment is non-analytical. **Random offers convert 0.9%**, the ops one-size-fits-all **low-cost offer converts 2.3%** but caps value at \$25.06 margin per convert, it sells the cheapest bundle to everyone.

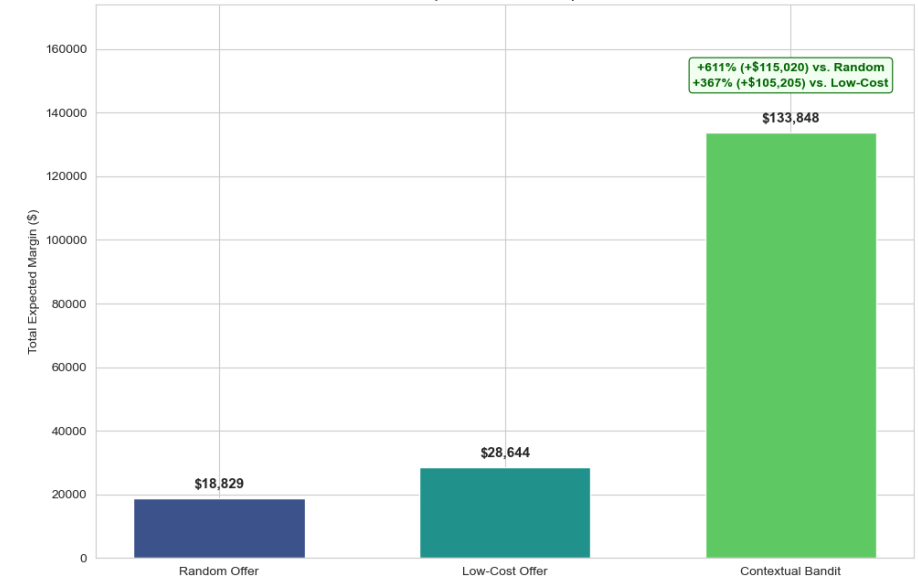


Resolution

A contextual-bandit recommender assigns each customer the valid bundle maximizing expected margin: $P(\text{convert}) \times \text{bundle margin}$, reaching **5.7% conversion** and **\$133,848 margin** on the official 50K test base.

Total expected margin: Official 50K validation base

Financial Impact: Total Expected Margin by Offer Strategy
(Official Test Set)



Dataset: final_test_validation_base

0.9% → 2.3% → 5.7%

CONVERSION: RANDOM → LOW-COST → BANDIT

\$2.68 vs \$0.57

MARGIN PER LEAD VS. LOW-COST

4.67× (+\$105,205)

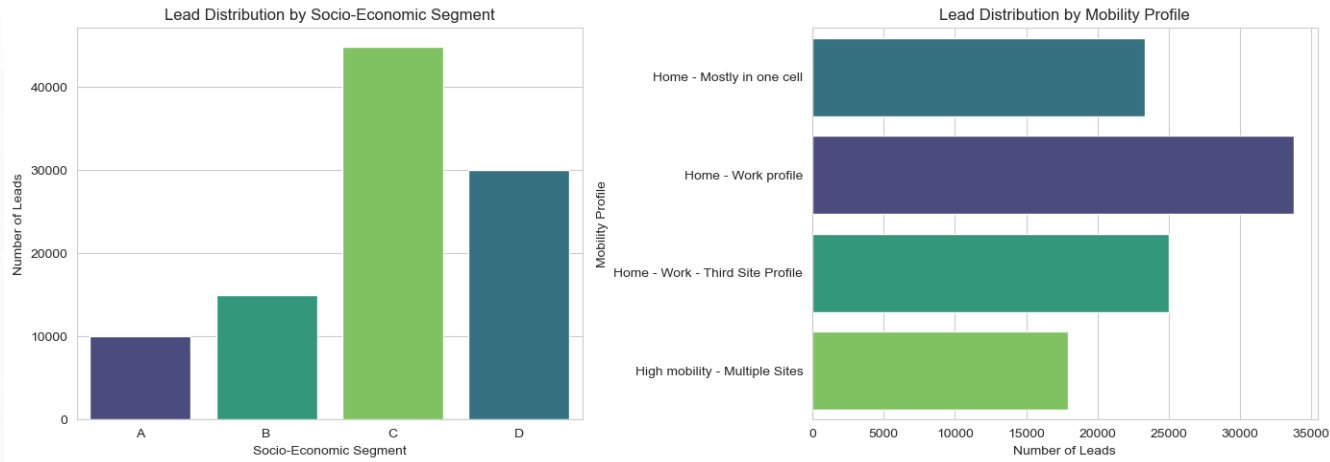
MARGIN LIFT VS. LOW-COST STRATEGY

7.11× (+\$115,020)

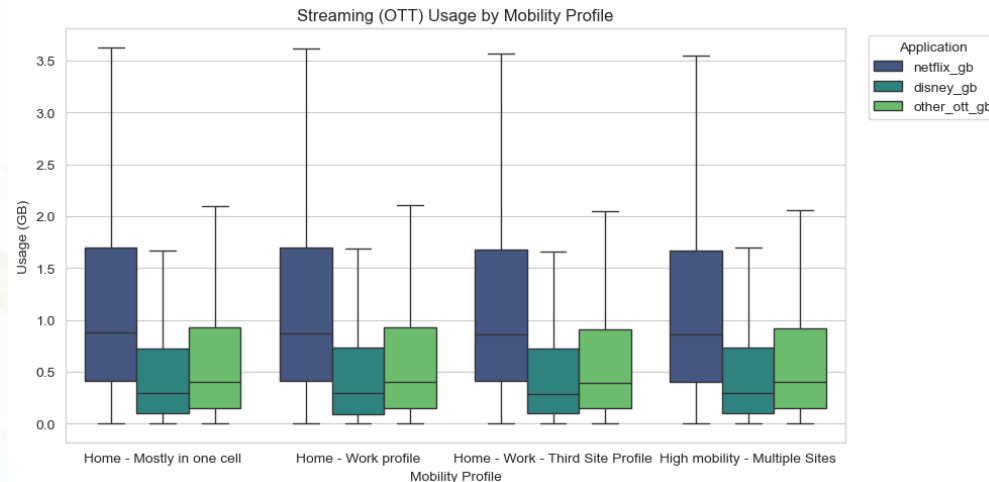
MARGIN LIFT VS. RANDOM BASELINE

The 100K lead base is not one market: Segment, mobility and streaming behavior define micro-segments with different willingness to pay

Who the leads are?



What they consume?



Affluence is scarce, not volume

Segment A is a minority of leads yet, **as the model later confirms (+2.83 log-odds), carries the highest conversion propensity.** Lead quality, not quantity, is the binding constraint.



Home time = Fixed-product need

Home-centric profiles concentrate streaming (Netflix, Disney, YouTube) consumption, **natural targets for high-speed internet**, Premium TV and OTT bundles. High-mobility users under-index on fixed-product need.

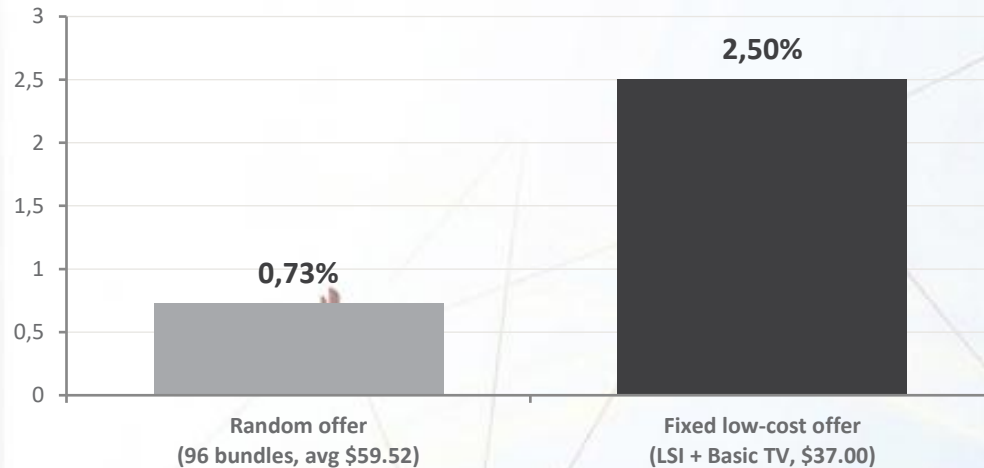


Three personas emerge

- **Affluent Home Streamer** (A/B, home-centric, heavy OTT: Premium bundles)
- **Value Commuter** (C/D, home-work: Lean offers)
- **Mobile-First Nomad** (high mobility: Hardest fixed-product fit).

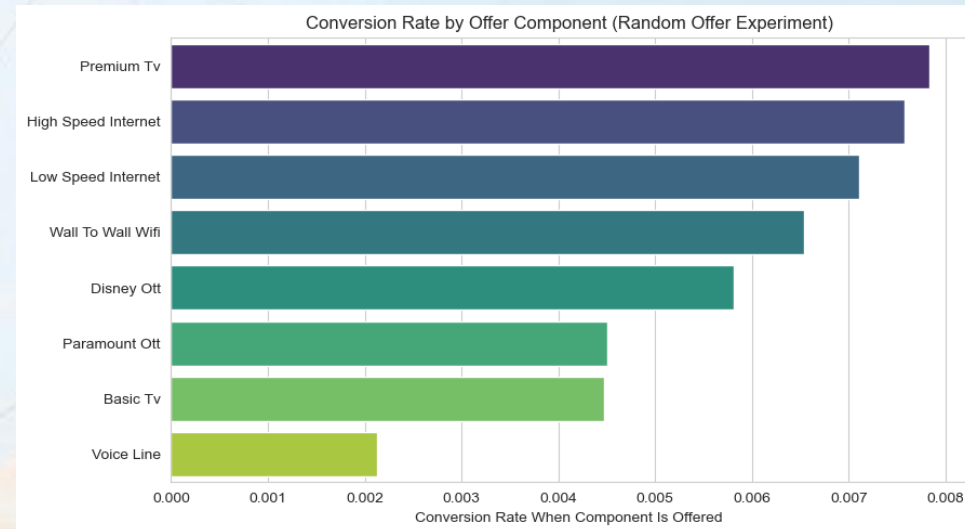
Two field experiments reveal the conversion levers: Customers reward simple, relevant bundles and punish irrelevant add-ons like Voice

Experiment outcomes (100K leads, 50K per arm)



A cheap, simple, universally understandable offer more than triples random conversion (0.73% → 2.50%, margin \$13.4K → \$31.4K). But it is a ceiling, not a strategy: It leaves premium demand unserved.

Acceptance rate when component is offered (random arm)



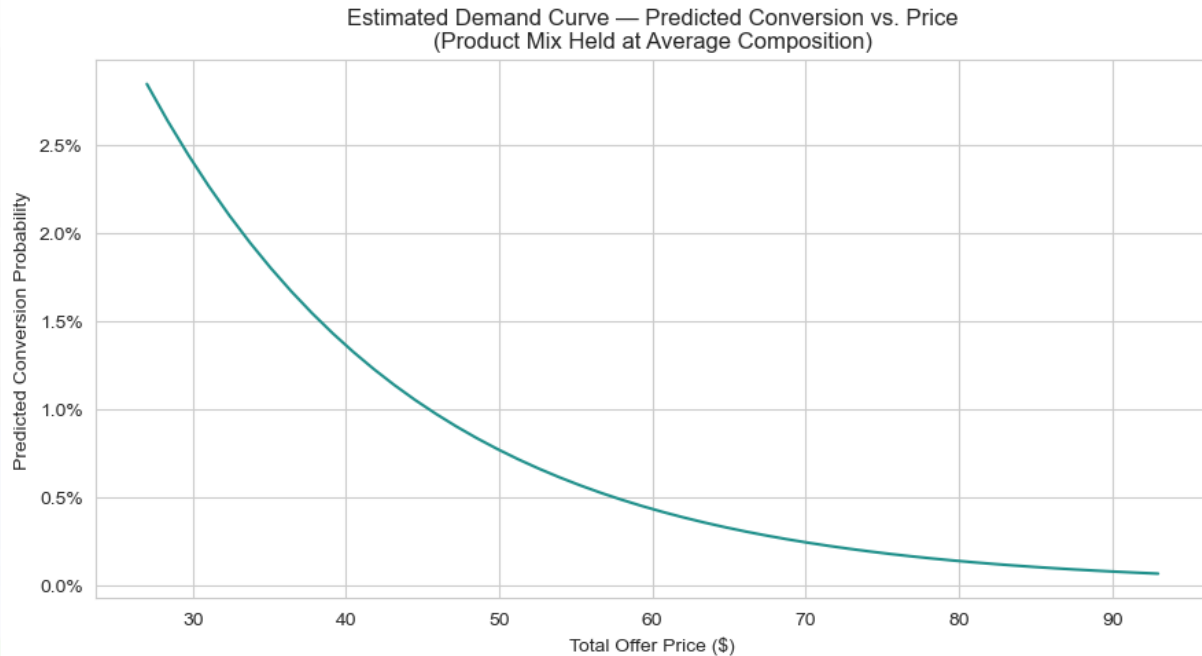
Premium components out-convert cheap ones even at higher prices, relevance beats price (Premium TV 0.78% ≈ High Speed 0.76% > Basic TV 0.45%). **Voice is a conversion killer** (0.21%; logit -1.31 log-odds, p<0.001): bundling it "for value" actively destroys calls.



Offer-customer match inference: Two demand families coexist in one base. Winning random bundles split into lean & cheap (Low Speed alone: 6.3% conv.) and rich & premium (High Speed + Premium TV + Disney + WiFi: 3.9% conv., top margin per customer at \$1.99). One fixed offer cannot serve both, personalization can.

Demand is elastic: Each extra dollar cuts conversion odds by 5.6%, but elasticity differs by profile. Price cuts only pay off in value segments

Estimated demand curve: Logit prediction, product mix held at average



Model: $\text{Logit}(\text{converted}) \sim \text{total_price} + \text{product-mix flags}$, fitted on the 50K random-offer arm. Price coefficient -0.0580 log-odds per \$1 (odds ratio 0.9437, $p = 0.028$). The randomized arm gives price variation the fixed offer cannot.

Magnitude: Demand slopes down

1

Each extra dollar charged multiplies conversion odds by 0.9437 (-5.6% per \$), holding product mix fixed. Statistically significant and consistent with standard demand theory.

Field confirmation: With an honest caveat

2

Random mix (\$59.52 avg) $\rightarrow$ low-cost offer (\$37.00): $+1.77$ pp conversion for a \$22.52 price cut ($3.41\times$ baseline). Caveat: the low-cost offer also strips add-ons, so part of the lift is mix simplification, not pure price.

Heterogeneity: Why one price does not fit all

3

Holding price constant, High Speed adds $+0.82$ log-odds ($p=0.023$) and Premium TV $+1.13$ ($p=0.07$): Premium features carry positive utility that offsets price for the right customers. Elasticity is profile-dependent, the case for personalized bundles over blanket discounts.



A contextual bandit: epsilon-greedy exploration over 96 valid bundles, scored by an interpretable logistic model, learns the policy in 10 waves

1. Context

3 categorical features one-hot + 9 numeric usage features standardized

2. Action space

96 valid bundles by construction: internet 2 × TV 3 × add-ons 16; invalid offers impossible

3. Policy

ϵ -greedy: Explore random valid bundle w.p. ϵ , else exploit model's best action

4. Reward model

Logistic regression $P(\text{convert} \mid \text{customer, offer})$, pick $\text{argmax } P \times \text{bundle margin}$

5. Learning loop

10 biweekly waves × 50K leads, retrain on ALL accumulated (context, action, reward)

Design choice: Why (every parameter has a business reason)?

ϵ : linspace 1.0 $\rightarrow$ 0.05 over 10 waves	Full exploration at cold start (no model exists), near-full exploitation once confident. Mirrors the biweekly campaign calendar.
Logistic regression (not GBM / NN)	Interpretable coefficients for marketing, scores $50K \times 96 = 4.8M$ pairs fast, rubric rewards a simple, fully-owned model.
class_weight = "balanced"	Conversions are rare (<3% early): Prevents collapsing to "never converts".
max_iter = 1000 · random_state = 42	Guaranteed solver convergence, fully reproducible learning curve.
Cumulative retraining (no sliding window)	Leads are i.i.d. draws from a stationary base, more data strictly reduces variance.
Objective: $\text{argmax } P(\text{convert}) \times \text{margin}$	Optimizes what the CMO cares about: Expected margin, not raw conversion.

Proof the loop learns: Conversion by training wave



Wave 1: 0.95% · \$18K margin $\rightarrow$ Wave 10: 5.40% · \$128.9K margin ($\epsilon = 0.05$). Wave-8 dip = sampling variance at $\epsilon = 0.26$.



RESULTS: THREE STRATEGIES COMPARED

On the official 50K validation base the bandit delivers \$133.8K margin, 4.67× low-cost, by steering every lead to a tailored high-speed bundle

Official test set: 50,000 leads, identical base for all three strategies

Strategy	Converted	Conv. rate	Total margin	Margin / convert	Margin / lead
Random offer	470	0.9%	\$18,829	\$40.06	\$0.38
Low-cost offer	1,143	2.3%	\$28,644	\$25.06	\$0.57
Contextual bandit	2,863	5.7%	\$133,848	\$46.75	\$2.68

Lift: vs. low-cost **+\$105,205 (4.67×, +367%)** - vs. random **+\$115,020 (7.11×, +611%)**

Why it wins: Top model drivers (log-odds coefficients)

+2.83

Segment A

+2.14

Home-centric mobility

-2.61

Voice line attached

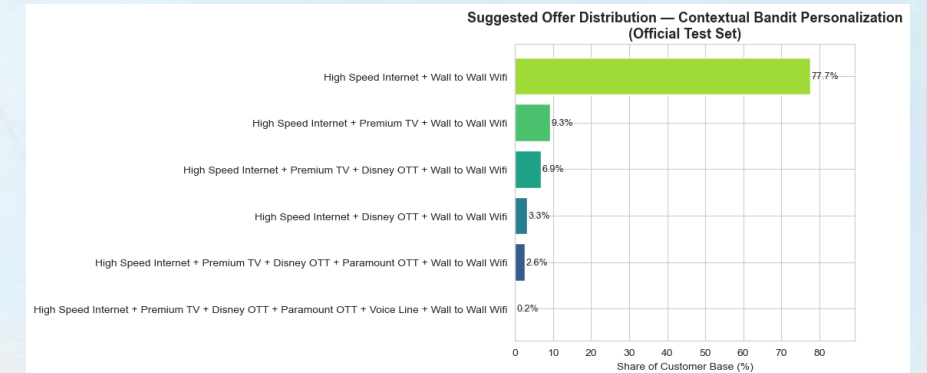
-2.17

Home-Work-3rd-site mobility

-1.57

iPhone mid-end device

Personalization proof: Offer mix chosen by the policy



The policy actively uses 6 of 96 valid bundles

- High Speed + WiFi 77.7%
- + Premium TV 9.3%
- + Premium TV + Disney 6.9%
- + Disney 3.3%
- + both OTTs 2.6%
- full stack 0.2%.

It never recommends Low Speed, Basic TV or standalone Voice, the exact opposite of today's ops strategy and always attaches the \$4 WiFi add-on, which raises perceived value more than it hurts price.

Concentration ≠ shortcut: Argmax of expected margin legitimately concentrates when one bundle dominates most of the customer space. The differential is who gets Premium TV / Disney added.

Scaling the recommender into an agentic CVM loop can compound the \$105K/wave lift while cutting campaign cycle time from weeks to days

High-level architecture: Four purposeful agents around an MLOps core



DATA LAYER

Usage / CDR feature store - CRM - Billing margins



MLOPS CORE

Feature pipeline - Model registry - Champion/Challenger serving API

AGENT LAYER

Offer Decision

Runs bandit policy per lead in real time, escalates low-confidence cases to exploration

Experimentation

Owens € budget & holdout design, every wave yields unbiased learning data

Insight & Narrative

Turns coefficient shifts into weekly CMO-Readable briefs

Monitoring / Drift

Watches calibration, triggers retraining or rollback



ACTIVATION

Contact-center CTI / predictive dialer · campaign manager



GUARDRAILS

- 96-action validity rules
- Pricing floors
- Contact policy
- Human approval for new offers

Processes transformed

1. **Lead scoring & offer assignment:** Monthly manual promo rules → real-time learned policy
2. **Experiment design:** Ad-hoc tests → automated ϵ / holdout management
3. **First-pitch call script:** Generic script → personalized opening (the case showed the first offer decides the call)
4. **Monitoring & retraining:** Manual reviews → continuous drift-triggered loop



MEASURING INCREMENTAL VALUE

Permanent randomized holdout (~5% of leads on the current low-cost rule) → **incremental margin per wave = $\text{margin}(\text{agentic}) - \text{margin}(\text{holdout, scaled})$** . Secondary KPIs: Cycle time design→launch, % offers auto-approved within guardrails, model calibration error. Baseline anchored on the measured +\$105K / 50K-wave lift.



HOW AI WAS USED TO DEVELOP THIS CASE

AI acted as a pair-programmer and thought partner under human direction: Every line of code, parameter and insight was reviewed and owned

PHASE

HOW AI WAS USED

HUMAN OWNERSHIP & VALIDATION



1. Exploration

Drafted matplotlib, seaborn boilerplate for distribution and multivariate charts to help identify descriptive insights from the base.

Defined which business question each chart must answer, discarded non-insightful graphs.



2. Modeling

Proposed candidate architectures (multiclass model, per-offer models, contextual bandit) and drafted implementation code.

Selected the bandit for its fit with the sequential 10-wave design and the “fully understood > complex” rubric, challenged every parameter (ϵ schedule, class weighting, action-space constraints).



3. Interpretation & storytelling

Stress-tested interpretations (log-odds vs. percentage points, mix-vs-price confound) and helped draft and structure this deck under the Pyramid Principle.

Wrote the final insights, verified every number against executed notebook outputs, define the storytelling and built the final deck.



Tools Used | Development Stack: VS Code + Opus 4.8 to draft code (Cell autocomplete blocked inside CLAUDE.md, all code-generation prompts logged). **Prompt Improvement:** Gem in Gemini. **Result Analysis:** Fable(assisted analysis, not automated).